JURY BRIEFING · ROUND 1 · 6 SEPTEMBER 2026

VayuChakra

Air Pollution and Weather Coupled Forecasting, Delhi NCR · MoES / NCMRWF

Every air quality forecast treats weather as something that happens *to* pollution. Over Delhi that is only half the story: the smog is thick enough to dim the sun, which shallows the layer it sits in, which concentrates it further. We close that loop, solve it, and then measure whether closing it helped.

12 / 12 forecast heads beat persistence on a winter they never saw	**8 / 8** coupled feedback magnitudes inside published Delhi ranges	**10 / 10** stations improved when held out of training entirely	**1,120** cells, 2.8 km over Delhi, live at vayuchakra.onrender.com

How to use this document

Sections 1 to 5 are what we say. Section 6 is what we admit before we are asked, which is the section juries actually reward. Section 7 is where we are going. Section 8 is a set of likely questions with answers we can defend, and it is the page worth reading twice before we walk in.

Live: vayuchakra.onrender.com **Code:** github.com/bindpratapsingh/VayuChakra **Tests:** 94 passing

Decisions logged: 63

1 The problem, in ninety seconds

SAY THIS

"Delhi's air is not a chemistry problem sitting on top of a weather problem. The two are one problem, and the coupling runs both ways. Almost every operational system models one direction. The Ministry's brief asks for both, and says leaving the second one out causes significant inaccuracies. That second direction is what we built."

1.1 The direction everyone models

Weather sets the volume that pollution occupies. On a winter night the ground cools faster than the air above it, the usual temperature profile inverts, and a lid forms over the city. The layer that emissions mix into can collapse from two kilometres in the afternoon to under two hundred metres before dawn. The emissions do not change. The air they dilute into shrinks by a factor of ten, and the concentration rises to match.

1.2 The direction that is hard, and is the brief

Aerosol is not a passive tracer being pushed around by the weather. It absorbs and scatters sunlight. Less energy reaches the ground, so the surface warms less, so the convection that grows the daytime boundary layer is weaker, so the layer stays shallow, so the wind through it slackens, so the same emissions end up more concentrated, which puts more aerosol in the column and blocks more sunlight still.

The last step feeds the first. That is why this cannot be evaluated once. It has to be solved. A forecast that treats meteorology purely as an input cannot represent it at all, and will systematically under-predict exactly the episodes that matter.

There is a second aerosol pathway and it is stronger for ozone. The same haze removes the ultraviolet that drives photochemistry. Published experiments separating the two find the radiation route moves ozone by 1 to 3 percent and the photolysis route by 10 to 12 percent. The brief names ground-level ozone explicitly, so we model the pathway that governs it.

1.3 And the worst of it is not made here

Through October and November, smoke from paddy stubble burning in Punjab and Haryana travels 200 to 400 km and arrives over a city that often cannot ventilate it. The brief is precise about what must be modelled: not the fires, but the interaction between the arriving plume and the inversion that decides whether it reaches breathing height or passes overhead.

The eleven clauses

We split the brief into eleven testable clauses and treat them as acceptance criteria. **Ten are met in full. One is partial by construction** and section 6.1 explains why in a way we can defend rather than apologise for.

2 What we built

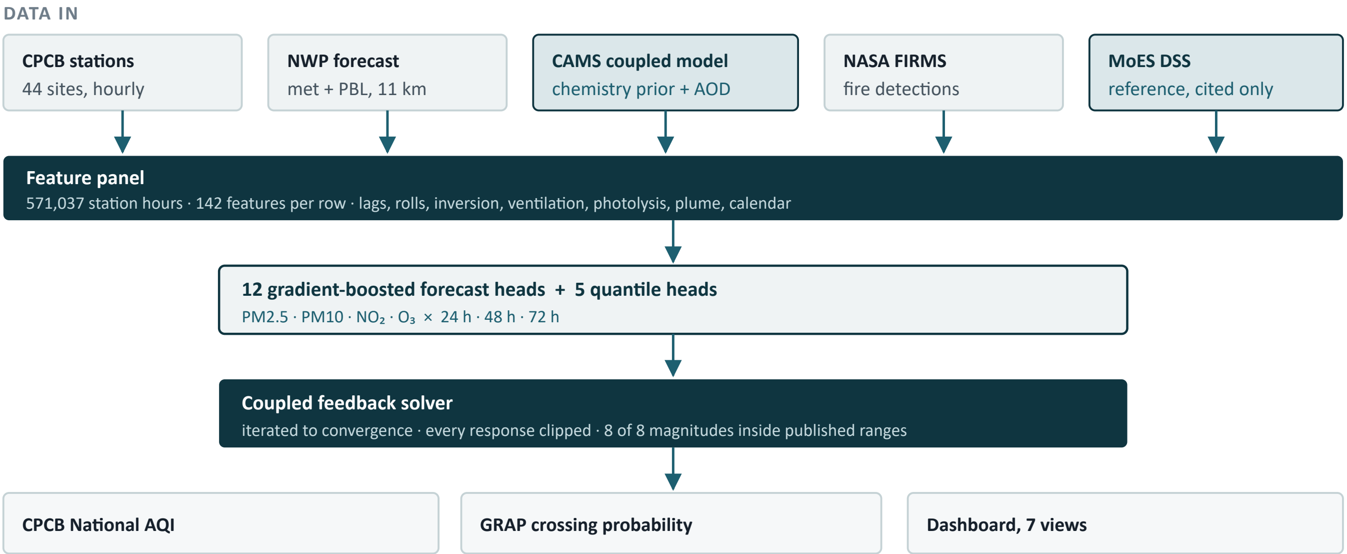

Figure 1. The two shaded inputs are third-party coupled models. CAMS supplies the chemistry prior and the aerosol optical depth the feedback runs on; the MoES DSS is the reference our plume is calibrated against. It is cited, never redistributed.

2.1 The six things a jury should remember

COMPONENT	WHAT IT DOES, AND WHY IT IS NOT STANDARD
Coupled solver	Runs the aerosol feedback loop to convergence rather than evaluating it once. Every one of eight responses is checked against published Delhi values the model was never fitted to.
Inversion tracker	The lid over the city as a first-class, displayed quantity: strength in kelvin, height in metres, and a time-height cross section. Most dashboards show a surface number and cannot show a lid at all.
Stubble plume	Satellite fire detections carried forward on the forecast wind. Reports *arrival*, not ignition: a thousand fires with the wrong wind are somebody else's problem that day, and the model says so.
Photolysis	MCM v3.3.1 clear-sky rates attenuated by aerosol. This is the pathway that governs ozone, and it is three to ten times stronger than the radiation route.
Uncertainty	Quantile heads giving the probability of crossing each GRAP trigger. A point forecast of 88 against a threshold of 90 tells a decision maker nothing.
Evidence view	Every validation number on screen, negative results included, each traceable to a file. Nothing on the site is typed in by hand.

3 How the AI actually predicts

SAY THIS

"The machine learning does one job: it maps a physical atmospheric state to a concentration. It does not invent the physics. The physics is computed separately, fed in as features, and then applied again on the output as an explicit solver. That split is deliberate, and it is why we can answer questions a pure statistical model cannot."

3.1 The models

PROPERTY	VALUE
Algorithm	Gradient-boosted decision trees (XGBoost)
Heads	12 point models: 4 pollutants × 3 horizons. Plus 5 quantile heads for PM2.5 at 24 h
Inputs	142 features per cell-hour
Training data	571,037 station-hours, 44 CPCB stations, Oct 2020 to Aug 2026
Held out	1 Nov 2025 to 28 Feb 2026, a whole winter, removed from training entirely
Baseline	Persistence: "tomorrow looks like today"

3.2 Why boosted trees and not a neural network

Three reasons we can defend. The data is tabular and moderately sized, which is the regime where boosted trees still beat deep networks on like-for-like tuning. They handle missing values natively, which matters when a station drops out for a week. And they are inspectable: we can say which features the model leaned on, which is how we caught a train-serve gap where three heads were being served a trained feature filled with nulls.

3.3 What the model actually learned

For PM2.5 at 24 hours the highest-ranked features are, in order: day of year, a winter flag, the cyclical day-of-year encoding, a monsoon flag, the 1-hour lag, the 72-hour rolling mean, and **the ventilation coefficient**. That last one is the physics earning its place. It is mixing depth multiplied by the wind through that depth, so it is the volume flux available to carry pollution away, and the model ranks it above most of the raw meteorology it was derived from.

Where the physics enters the machine learning

Inversion strength, lid height, mixing depth, ventilation coefficient and the photolysis rates are all **computed** by physical modules and then handed to the model as features. The model is not asked to rediscover atmospheric stability from temperature and pressure. It is asked to map a stability that we calculated onto a concentration that a station measured, which is a far easier and far better-posed problem.

3.4 And then physics is applied again, on the output

The trained model produces an uncoupled forecast. The solver then takes that number and asks a narrower question with a published answer: given this much aerosol, how much less sunlight reaches the ground, how much shallower does the layer become, and what does that do to the concentration that produced it. Because the last step feeds the first it is iterated to convergence, with every response clipped to a physical range and any cell that will not converge falling back to the uncoupled answer and being counted.

This is what lets us answer **what would ozone be if the aerosol halved**. No purely statistical model can, because that atmosphere is not in its training data. A mechanism can. Ours gives +12.79 percent against a published Delhi figure of +25 percent: the same sign, within a factor of two, and it was never fitted to that number.

3.5 How we know it is not memorising

TEST	WHAT IT REMOVES	RESULT
Held-out winter	Four months of the target season, entirely	**12 of 12 heads beat persistence**
Leave one station out	A whole station's data from training, then predicts it	**PM2.5 10 of 10, mean +31.6%**
Ozone counterfactual	Asks about an atmosphere not in the data	**Same sign, within 2× of published**
Quantile calibration	Checks the stated uncertainty is real	**80% interval covers 75.6%**

Leave-one-station-out is the one that matters most for a gridded product: we serve 1,120 cells and about forty contain an instrument, so the real question is whether it works where there is no monitor. Persistence still gets that station's own history; the model does not.

4 The loop, which is the differentiator

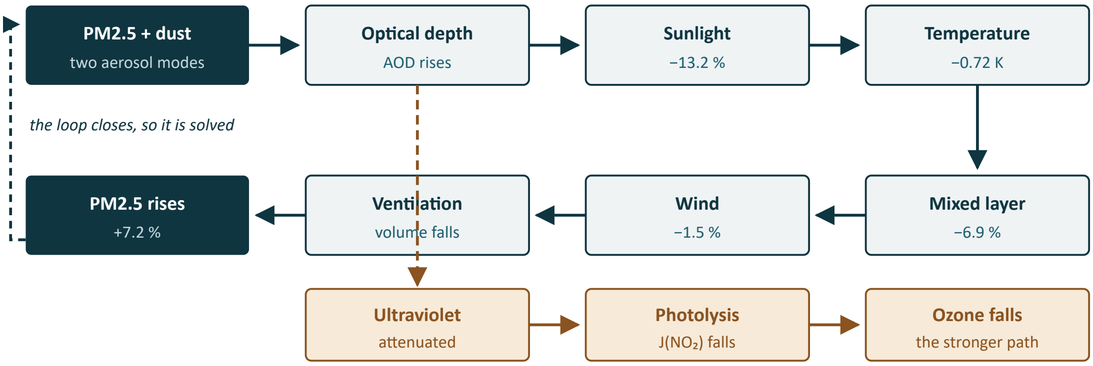

Figure 2. Percentages are measured values from the shipped run, under haze. The amber branch is the photolysis pathway, which is the one that governs ozone.

4.1 Every magnitude checked against something we did not fit to

RESPONSE UNDER HAZE	OURS	PUBLISHED	RESULT
Sunlight reaching the surface	−13.24%	5 to 35%	inside
Daytime temperature	−0.72 K	0.1 to 2.5 K	inside
Boundary layer height	−6.87%	5 to 35%	inside
Surface wind	−1.45%	0.5 to 6%	inside
PM2.5 amplification	+7.19%	2 to 30%	inside
PM10 amplification	+5.77%	1 to 25%	inside
NO_2 amplification	+16.32%	2 to 35%	inside
Ozone response	−0.04%	must be negative	correct sign

Converged in 5 iterations, zero cells fell back, 70,449 cell-hours in the high-aerosol daylight regime the published figures were measured in. The gate is a hard test in the suite, not a report: a physically absurd result fails the build.

5 Results we can put on a slide

CLAIM	NUMBER	WHAT IT MEANS
Forecast skill	+15.0% to +28.8%	All 12 heads beat persistence on a held-out winter. The margin grows with lead time, which is the right shape.
Where there is no monitor	+31.6%	PM2.5, 10 of 10 stations improved when removed from training entirely.
Coupling magnitudes	8 of 8	Inside published Delhi ranges never used in fitting.
Ozone counterfactual	+12.79%	For a 50% aerosol cut, against a published +25%. Same sign, within 2×, never fitted.
Stubble plume	r = +0.60	Against the MoES DSS daily attribution over 54 days. Peak 27.7 against their 38.0 μg/m³.
GRAP decision support	+0.34 to +0.39	Brier skill against climatology for crossing each GRAP trigger.
Uncertainty honesty	75.6%	Measured coverage of a nominal 80% interval. Scored "well calibrated".

The comparison we are careful with

Against the MoES DSS over Oct 2021 to Feb 2022 at 24 hours, our RMSE is 60.81 against their 98.78 and persistence at 93.98. **We do not present this as beating the Ministry's system.** Our hindcast is driven by archived forecast meteorology, which removes most of the unfair advantage an earlier reanalysis version had, but not all of it: the archive holds the best available run for each hour, which is a short lead, while the DSS was scored 24 to 72 hours ahead. What the table supports is that our statistical layer maps meteorology to PM2.5 competitively. That is the sentence to use.

6 What we admit before we are asked

> SAY THIS
>
> *"There are four things we would fix with more time, and one we cannot fix at all. We would rather tell you what they are than have you find them."*

6.1 We do not run WRF-Chem, and it is not an effort problem CLAUSE 1 PARTIAL

The brief says to leverage advanced weather-chemistry models such as WRF-Chem *or similar open-source coupled frameworks*. We consume CAMS, which is ECMWF's operational global model with online chemistry and interactive aerosol, and we use the MoES WRF-Chem DSS as our calibration reference. We do not run a chemical transport model ourselves.

We researched whether we could, and the numbers are decisive rather than discouraging:

CONSTRAINT	WHAT WRF-CHEM NEEDS
Compile	Single-threaded to fit 16 GB, taking up to 4 hours of a 6-hour CI limit before any physics runs
Runtime	12 to 18 hours for a 72-hour forecast on 4 cores at 10 km
Memory	10 to 12 GB peak with the chemical solver's intermediates, with no swap available
Data	Roughly 2 GB of boundary conditions per run, plus a district emissions inventory we do not have

CMAQ and CAMx fail the same way: 5 to 15 GB of meteorological files exhaust ephemeral disk before the model starts.

This is the honest framing: running WRF-Chem is a national modelling centre's programme, not a hackathon's. What we built instead is a coupled surrogate whose every magnitude is checked against the literature, plus a route to a genuine mechanism described in section 7.

6.2 The coupling does not improve forecast error NEGATIVE RESULT

Measured, and we report it rather than bury it. Overall the coupled run is **0.11 percent worse** than the uncoupled one. Split by regime it is more interesting: it is 0.35 percent *better* in high-aerosol conditions and slightly worse in clean and well-ventilated ones.

> **How to answer this if a juror presses**
>
> "It helps exactly where the physics says it should act, and does nothing where the physics says it should not. That is the shape a correct but weak mechanism produces. The trained model has already learned much of this effect through its radiation and boundary-layer features, so the explicit solver is adding little on top. What the solver buys us is not accuracy, it is the ability to answer counterfactual questions the statistical model cannot, and to show a decision maker *why* a number is what it is."

6.3 Problems we hit and what they cost

PROBLEM	WHAT WE DID
The free host has 512 MB; the pipeline needs 1.1 GB	Measured that it does not fit at *any* grid resolution, so coarsening would have cost resolution and bought nothing. A GitHub Actions runner with 16 GB now runs the real pipeline every six hours and commits the result, which the site serves. We give up freshness, not resolution or physics.
Our optical depth used fine particles only	Found via research that this underestimates the column by 60 to 75 percent in the pre-monsoon dust season. Rebuilt with two aerosol modes carrying their own optical properties. Fixed this week.
Our NO_2 chemistry assumed clear-sky sink behaviour	Delhi haze shifts nitrogen loss from photolysis to aerosol-surface reactions. Corrected, and it made our headline NO_2 number *smaller*, from 20.2 to 16.3 percent. We changed it anyway.
A validation we planned did not work	We intended to check our ultraviolet attenuation against satellite UV data. Three measurements showed that product carries a climatological aerosol and has no daily signal to compare against. We recorded the negative result and used a stronger test instead.

6.4 Built but not yet promoted

A two-layer atmospheric model that lets stubble smoke sit in a residual layer overnight and mix down during the morning, rather than crossing a height threshold. It reproduces the observed 07:00 to 10:00 fumigation peak from a mass budget. It is written, tested and shipped as a scoreable option, but the shipped default is still the simpler scheme until we have scored both against the DSS. We would rather promote it on evidence than on the fact that it is newer.

7 Way forward

#	WHAT	EFFORT	WHY IT MATTERS
1	**Score the two-layer model against the DSS**	Half a day	Decides whether the residual-layer scheme replaces the threshold. Turns a built feature into a defended one.
2	**A 1D column chemistry model**	2 to 3 days	The route to closing clause 1. A single-column model with an explicit chemical mechanism fits in CI comfortably and can predict responses to emission changes it has never seen. It also fixes something CAMS gets structurally wrong: at 40 km its grid dilutes Delhi's nitrogen oxides across 1,600 km^2, which flips the modelled ozone regime.
3	**Ozone regime map from satellite**	1 to 2 days	Formaldehyde to nitrogen dioxide ratio from TROPOMI tells us whether ozone here is limited by hydrocarbons or by nitrogen oxides. It would make our ozone forecast regime-aware without needing hydrocarbon measurements we do not have.
4	**Ablate the three newer species**	3 hours	PM10, NO_2 and ozone coupling are bounded by literature but not yet scored against observations the way PM2.5 has been.
5	**Strict lead-matched DSS comparison**	1 to 2 days	Removes the last caveat on our headline comparison.

What we need, and it costs nothing

Item 2 needs open emissions data, and it is available: EDGAR v8.1 at 10 km under a Creative Commons licence, and SMoG-India, which has Indian brick-kiln and biomass-burning profiles from IIT Bombay. A free Copernicus account would add a third option. No paid compute is required for anything on this list.

7.1 If we are asked "what would you do with three more months"

Resolve the vertical properly rather than in two layers, add hydrocarbon chemistry so ozone is modelled on both of its limiting pathways rather than one, and assimilate station observations into the initial state instead of only using them as training targets. Those three would take this from a well-validated surrogate to something a forecaster could run operationally.

8 Likely questions, and answers we can defend

Is this just a machine learning model with a physics story on top?

No, and the test is that we can answer questions no fitted model can. Ask ours what ozone does if aerosol halves and it says +12.79 percent, against a published Delhi figure of +25, for an atmosphere that appears nowhere in its training data. It gets that from a mechanism. A pure statistical model can only interpolate weather it has already seen.

You say the coupling does not improve accuracy. So why include it?

Because the brief asks for it, because it is correct where it should act and inert where it should not, and because accuracy is not the only thing a decision maker needs. The solver is what lets us say *why* a forecast is high and what would change it. We also report the negative number rather than hiding it, which is the reason you can trust the positive ones.

How do we know the model is not just memorising Delhi's seasons?

Three ways. We removed an entire winter from training and it still beat persistence on all twelve heads. We removed whole stations and predicted them blind, improving on ten of ten for PM2.5. And the counterfactual above asks about an atmosphere that has never existed in the record.

Why not WRF-Chem?

We measured it rather than assumed. Compilation alone consumes up to four hours of a six-hour window, the run needs twelve to eighteen hours, and peak memory is ten to twelve gigabytes against sixteen available with no swap. It is not a matter of more effort. Section 7 item 2 is our route to a genuine chemical mechanism that does fit.

Your dashboard says the forecast is precomputed. Is it live or not?

The pipeline runs every six hours on a 16 GB runner and the result is deployed automatically. What you see is at most six hours old and it says so on every view. No operational forecast system re-solves its physics when you load a page; weather models run on a cycle and serve the most recent one. We do the same thing and we label it.

How is this different from the ward-level AQI apps that already exist?

Those answer "should I go outside", which is a good question and a different product. This is built around the vertical structure of the atmosphere and regional transport, because that is what the brief asks about. We show the lid over the city, where the smoke is coming from, and the probability of crossing a GRAP trigger, which is what an official actually decides on.

What is the weakest part of the system?

Chemistry is solved in a single well-mixed layer. Vertical structure is diagnosed and drawn but not resolved, so a plume arriving above the inversion is handled by a rule rather than by explicit mixing. We have built the two-layer replacement and it reproduces the observed morning peak; we have not yet scored it, so we have not yet switched to it.

The one sentence to close on

"We built the loop the brief asks for, checked all eight of its magnitudes against published values we never fitted to, and then measured whether it made the forecast better. It did not, by much, and we are telling you that because a system that only reports its wins has not been validated."